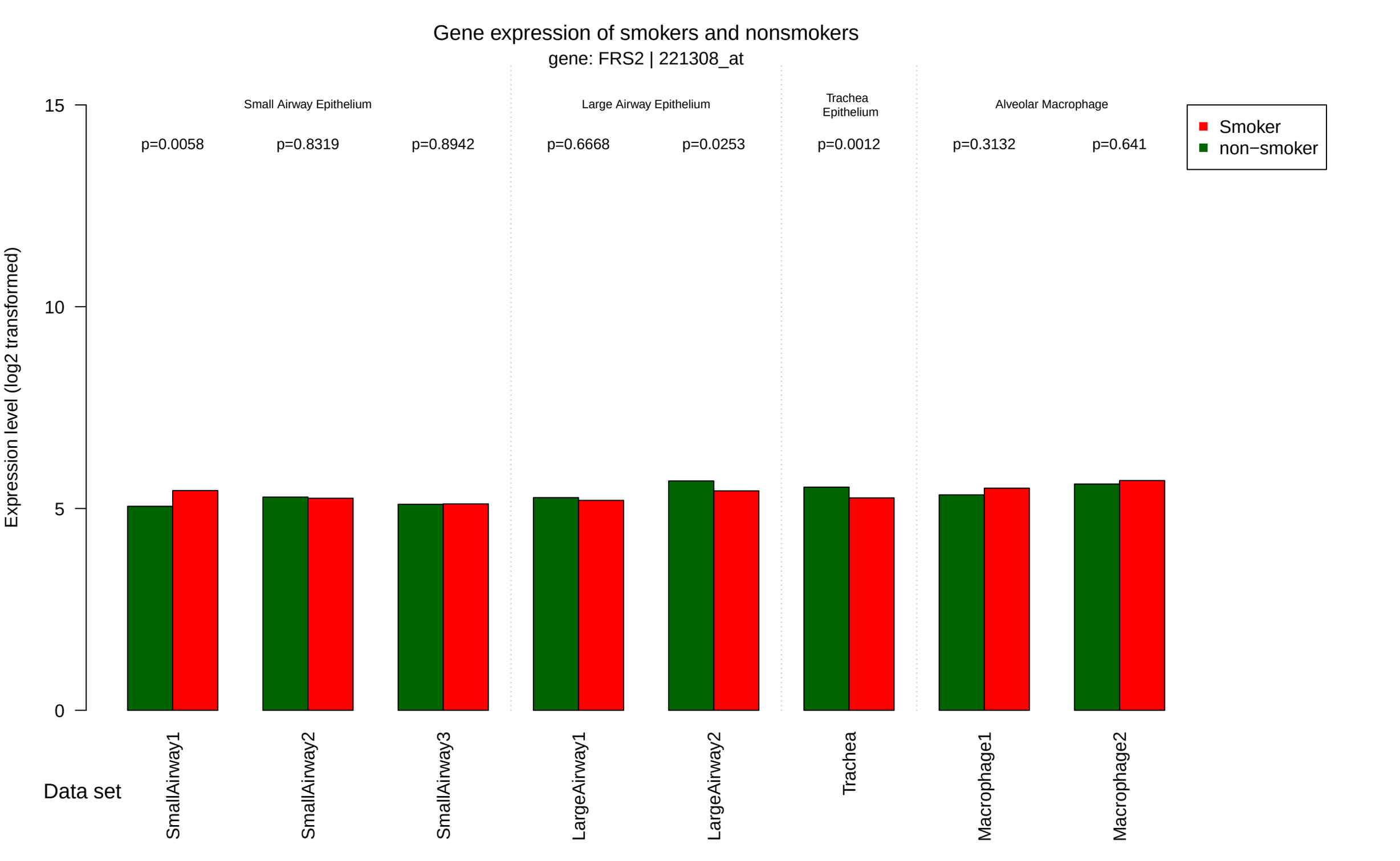
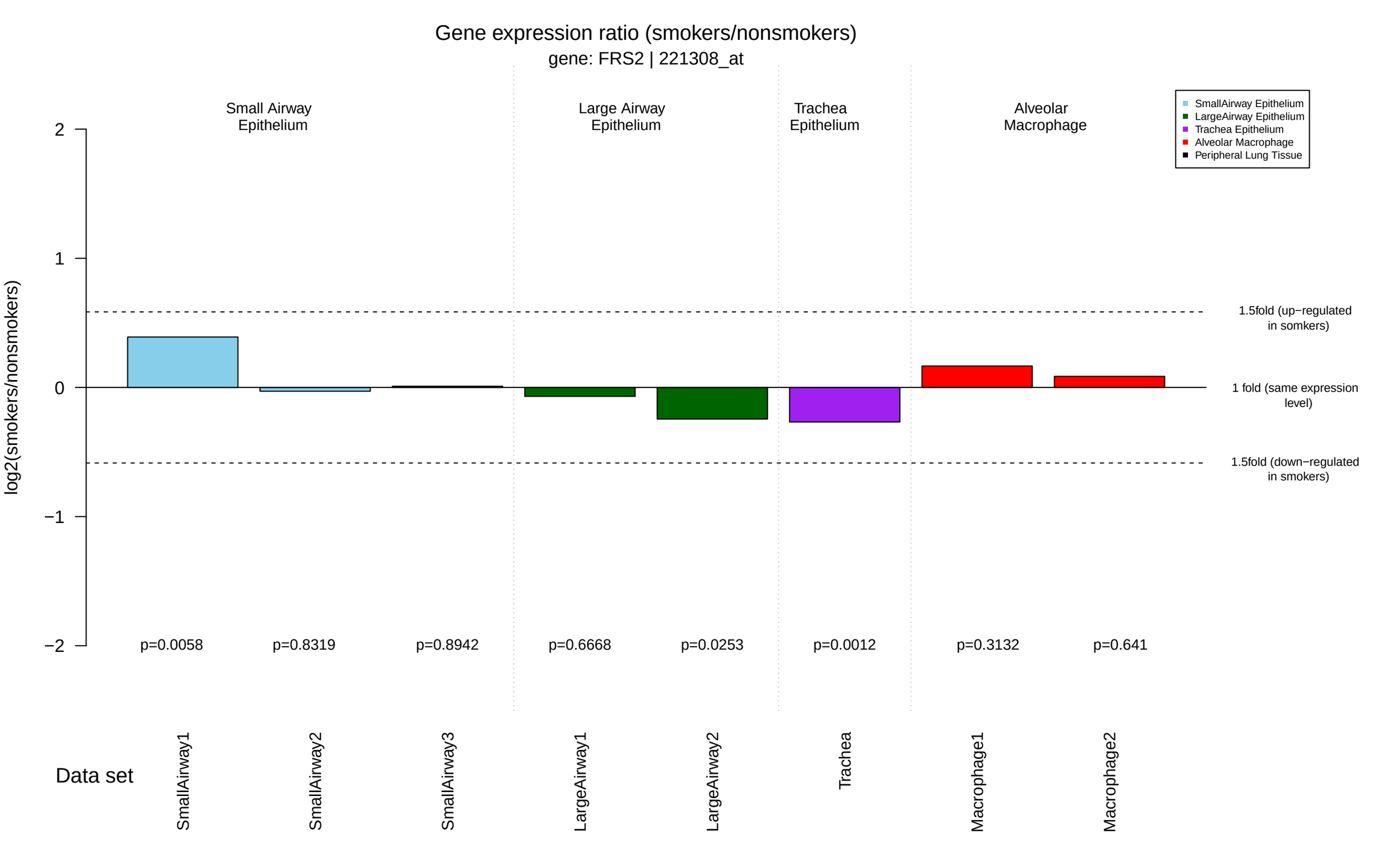
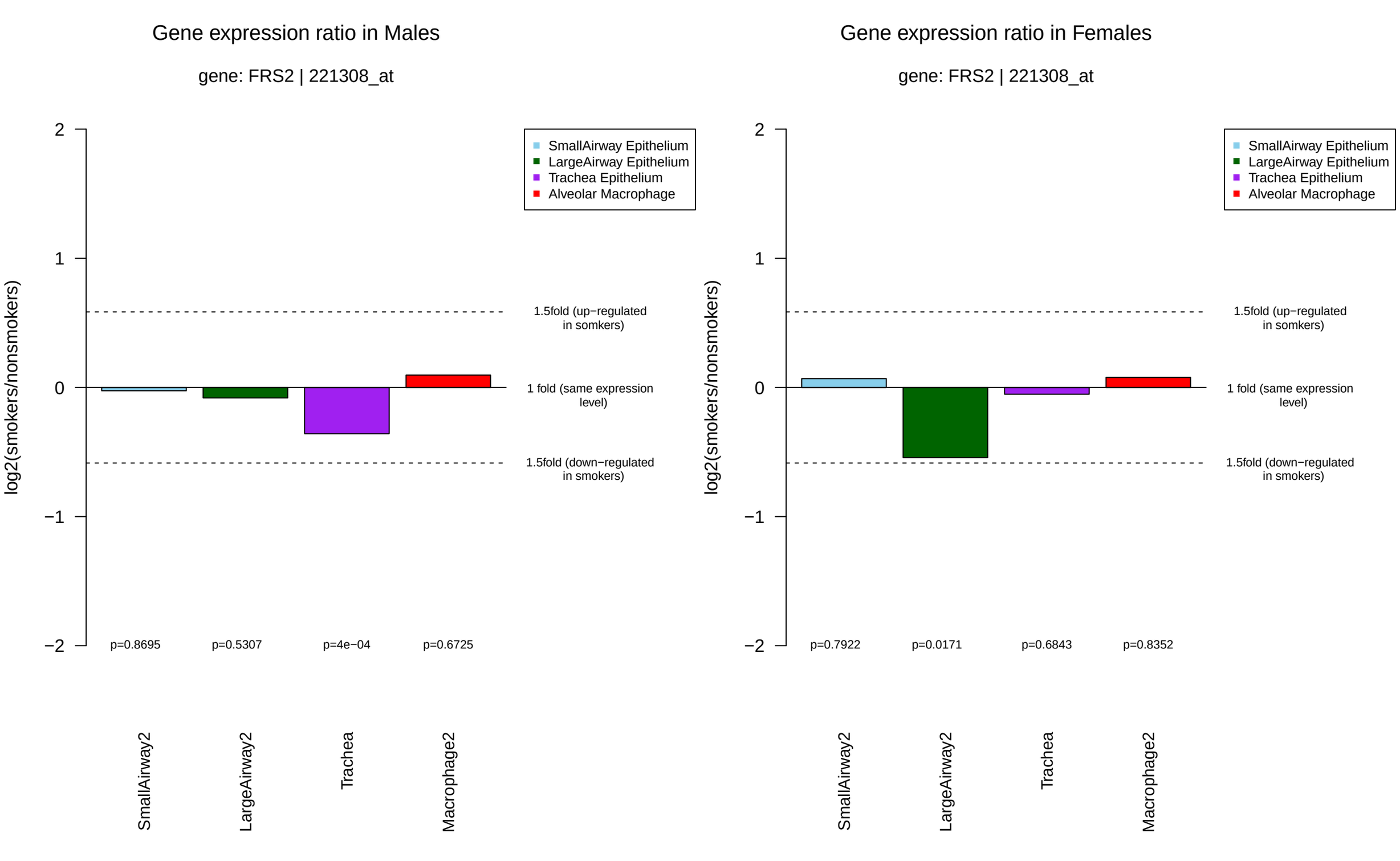
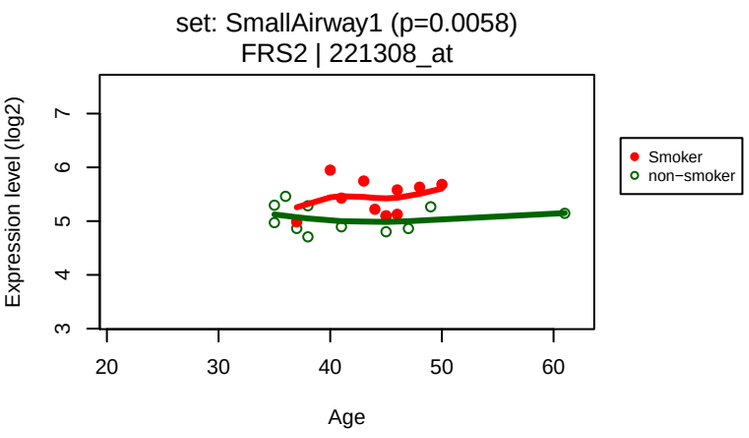
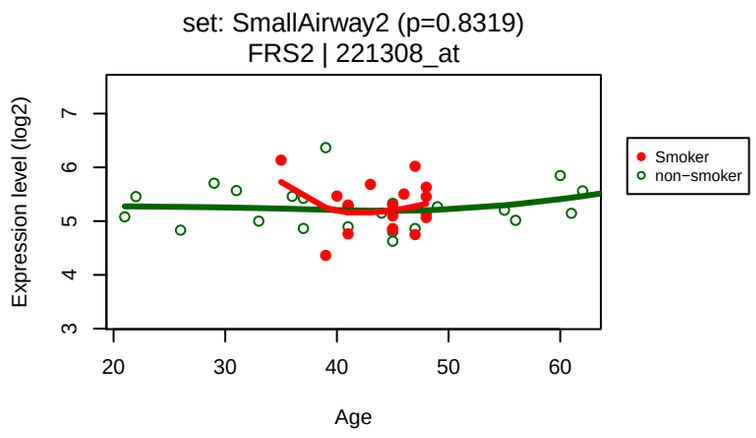
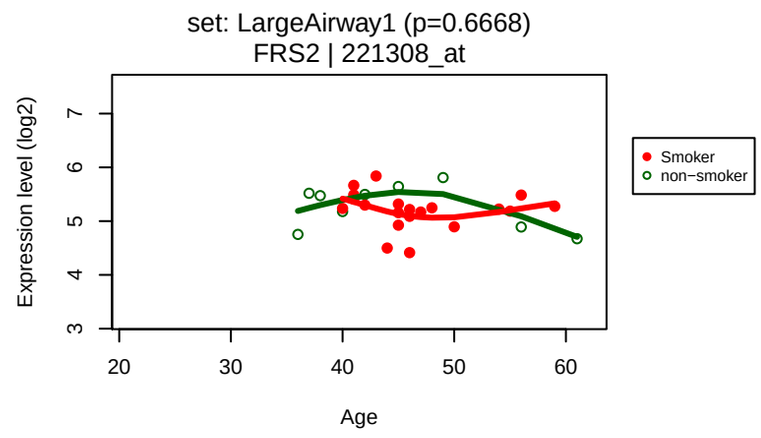

Gene expression of smokers and nonsmokers
gene: FRS2 | 221308_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0058
p=0.8319
p=0.8942
p=0.6668
p=0.0253
p=0.0012
p=0.3132
p=0.641
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

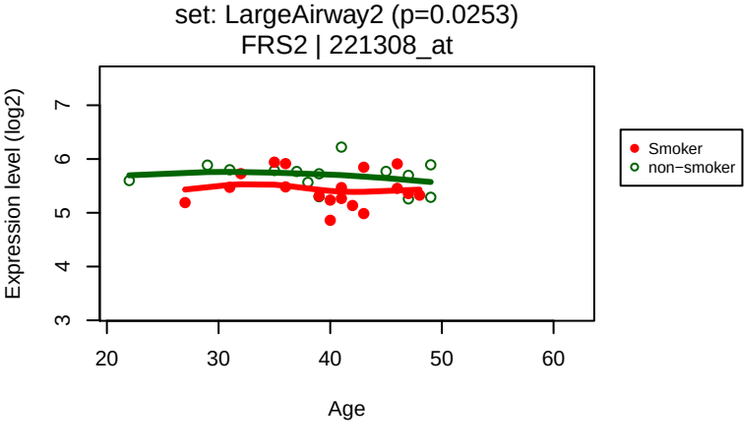

Gene expression ratio (smokers/nonsmokers)
gene: FRS2 | 221308_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0058
p=0.8319
p=0.8942
p=0.6668
p=0.0253
p=0.0012
p=0.3132
p=0.641
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

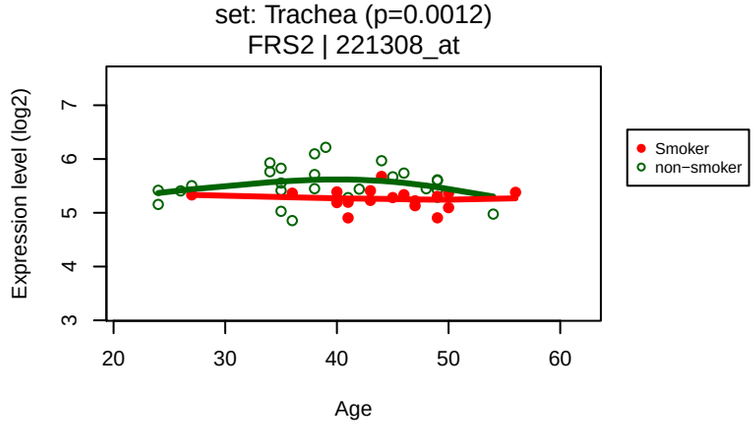

Gene expression ratio in Males
gene: FRS2 | 221308_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8695
p=0.5307
p=4e−04
p=0.6725
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FRS2 | 221308_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7922
p=0.0171
p=0.6843
p=0.8352
SmallAirway2
LargeAirway2
Trachea
Macrophage2

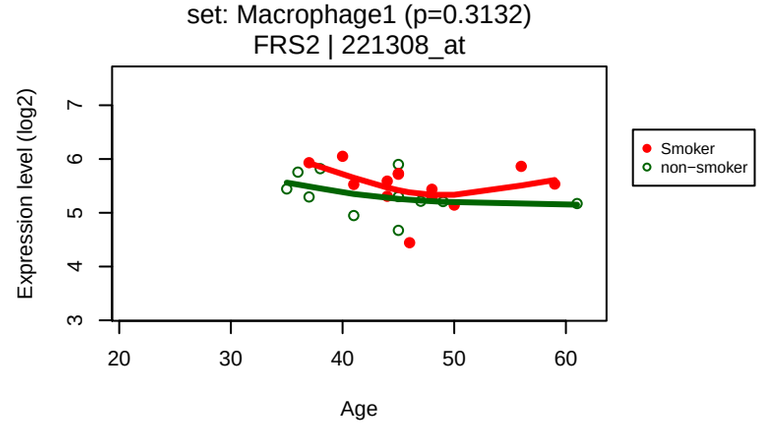

set: SmallAirway1 (p=0.0058)
FRS2 | 221308_at
Expression level (log2)
Age
Smoker
non−smoker

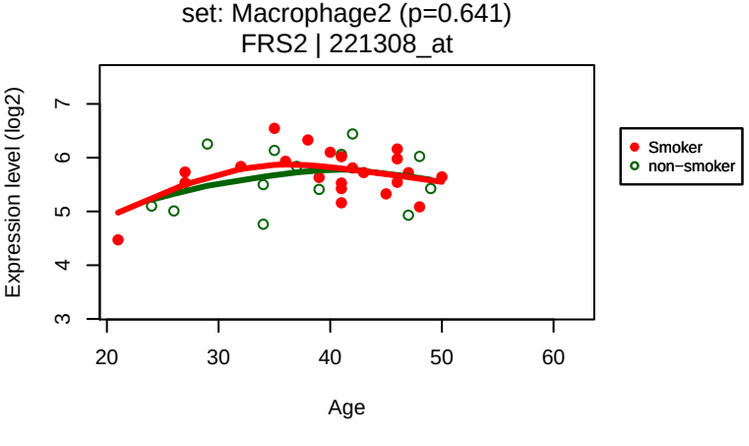

set: SmallAirway2 (p=0.8319)
FRS2 | 221308_at
Expression level (log2)
Age
Smoker
non−smoker

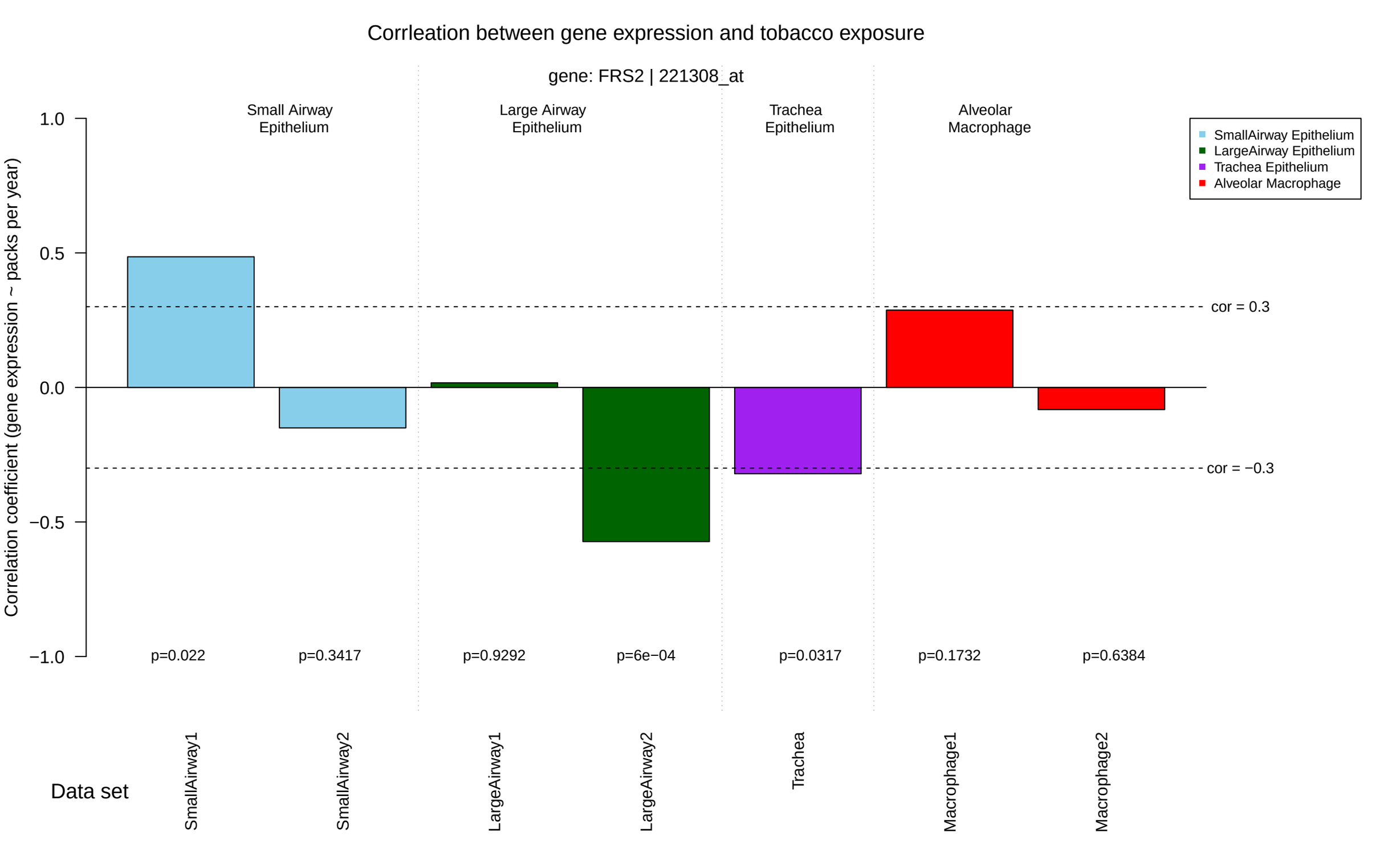

set: LargeAirway1 (p=0.6668)
FRS2 | 221308_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0253)
FRS2 | 221308_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0012)
FRS2 | 221308_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3132)
FRS2 | 221308_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.641)
FRS2 | 221308_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FRS2 | 221308_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.022
p=0.3417
p=0.9292
p=6e−04
p=0.0317
p=0.1732
p=0.6384
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2